

Bhav-Net: Semantic Analysis of Multilingual Antonym-Synonym Detection Using Dual-Encoder Graph Transformers

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Abstract

This paper presents Bhav-Net, a multilingual dual-encoder architecture that extends traditional semantic relationship detection to cross-lingual antonym-synonym identification. Building upon the foundational dual encoder framework, Bhav-Net integrates language-specific fine-tuned BERT models with graph transformer networks to model word relationships through dual projections in semantic space. The architecture employs separate synonym and antonym projection branches, enabling the model to learn distinct semantic patterns for contrasting relationships across seven languages: German, French, Spanish, Italian, Portuguese, Dutch, and Russian. Our approach leverages graph-based representations where word pairs form two-node graphs, allowing transformer convolutions to capture higher-order semantic dependencies. The model uses fine-tuned base BERT models to create initial graph representations for accelerated learning. Experimental evaluation demonstrates robust cross-lingual performance, with the dual-space projection mechanism proving particularly effective at distinguishing antonymous relationships from synonymous ones across diverse linguistic contexts. The margin-based loss function guides semantic space organization, creating interpretable representations that generalize effectively across linguistic boundaries.

Keywords: multilingual NLP, antonym detection, graph neural networks, dual encoder, semantic analysis, cross-lingual processing

1 Introduction

Detecting semantic relationships between words is a core challenge in natural language processing. While identifying synonyms is relatively straightforward, antonym detection presents unique difficulties. Antonyms share semantic domains but express opposite meanings - "hot" and "cold" both relate to temperature but represent opposing concepts. This creates a puzzle for computational models: antonyms often appear in similar contexts, making them hard to distinguish from synonyms using traditional approaches.

Most existing work focuses on single languages, limiting applicability in our multilingual world. Recent advances in transformer-based language models offer new possibilities for cross-lingual understanding, but few approaches explicitly model the dual nature of semantic relationships across languages.

We introduce Bhav-Net, a multilingual architecture that tackles these challenges head-on. Our approach builds on the dual encoder concept but extends it significantly for multilingual

contexts. The key insight is simple: synonyms and antonyms require different semantic spaces to be properly understood. We create separate projection branches for each relationship type and use graph transformers to capture the relational patterns that exist across languages.

Bhav-Net makes three main contributions. First, we design a dual-space projection mechanism that learns distinct patterns for synonym and antonym relationships. Second, we use graph representations with transformer convolutions to model higher-order dependencies between word pairs. Third, we employ language-specific fine-tuned BERT models as initial encoders, providing strong foundational representations for accelerated learning across seven languages.

Our results show that this architecture generalizes well across diverse languages, suggesting that the fundamental patterns of semantic opposition transcend linguistic boundaries while respecting language-specific characteristics.

2 Related Work

Early antonym detection relied on lexical resources like WordNet [6]. While effective for covered vocabulary, these methods struggled with new words and cross-lingual scenarios. Distributional approaches using word embeddings [5] faced the fundamental challenge that antonyms often appear in similar contexts, leading to high similarity scores despite opposite meanings.

Recent neural approaches have attempted various solutions. Graph neural networks have shown promise for modeling relational data [3], particularly for linguistic tasks requiring complex relationship modeling [4]. The transformer architecture’s adaptation to graphs [2] combines attention mechanisms with structural modeling capabilities.

Multilingual language models like mBERT [1] have demonstrated cross-lingual semantic understanding. However, most existing multilingual approaches for semantic relationship detection do not explicitly model the contrastive nature of antonymous relationships.

The dual encoder concept has shown promise for separating different semantic relationships, but previous work lacked sophisticated graph-based modeling and comprehensive multilingual evaluation. ConceptNet [7] provides valuable multilingual semantic knowledge but requires specialized architectures to effectively leverage this information for antonym detection.

Our work builds on these foundations by combining dual-space modeling with graph transformers in a unified multilingual framework, specifically designed to capture the nuanced differences between synonymous and antonymous relationships across languages.

3 Preliminaries

3.1 Problem Definition

Given a multilingual word pair dataset $\mathcal{W} = \{(w_i^1, w_i^2, y_i, l_i)\}_{i=1}^N$ where w_i^1 and w_i^2 are words, $y_i \in \{0, 1\}$ indicates the relationship (1 for antonym, 0 for non-antonym), and l_i specifies the language. Our goal is to learn a function $f : (w_i^1, w_i^2, l_i) \rightarrow \{0, 1\}$ that accurately predicts antonymous relationships across different languages.

3.2 Dual Encoder Foundation

The dual encoder approach recognizes that different semantic relationships require different representational spaces. Traditional approaches struggle because they treat all semantic relationships uniformly, leading to poor separation between synonyms and antonyms. By creating dedicated projection spaces, we can learn specialized patterns for each relationship type.

3.3 Graph-Based Word Pair Modeling

We represent each word pair as a simple two-node graph. This seemingly simple representation becomes powerful when combined with transformer convolutions, allowing the model to perform message passing between word representations and capture relational dependencies that go beyond simple similarity measures.

4 Architecture and Training Process

4.1 Overall Architecture

Bhav-Net consists of four main stages: (1) Language-specific BERT encoding, (2) Dual-space projection, (3) Graph transformer processing, and (4) Classification with margin-based optimization. This pipeline is designed to respect both universal semantic principles and language-specific characteristics.

4.2 Language-Specific BERT Encoding

We use specialized BERT models for each language rather than a single multilingual model. This design choice recognizes that different languages have unique morphological, syntactic, and semantic patterns that are better captured by dedicated models.

For each language, we employ the following models:

- German: `dbmdz/bert-base-german-cased`
- French: `camembert-base`
- Spanish: `dccuchile/bert-base-spanish-wwm-cased`
- Italian: `dbmdz/bert-base-italian-cased`
- Portuguese: `neuralmind/bert-base-portuguese-cased`
- Dutch: `GroNLP/bert-base-dutch-cased`
- Russian: `DeepPavlov/rubert-base-cased`

Each word w is encoded independently to obtain embeddings $\mathbf{x}_w \in \mathbb{R}^{768}$, preserving individual semantic characteristics while enabling subsequent relational modeling.

4.3 Dual Projection Mechanism

The core innovation of Bhav-Net lies in explicitly modeling the fact that synonyms and antonyms require different semantic perspectives. Think of it this way: when we hear "hot" and "warm," we recognize their similarity in the domain of temperature. But when we hear "hot" and "cold," we recognize both their shared domain and their opposition within that domain.

4.3.1 Synonym Space Projection

The synonym projection branch learns to position semantically similar words close together:

$$\mathbf{x}_{syn} = \text{Dropout}(\text{ReLU}(W_{syn}\mathbf{x} + \mathbf{b}_{syn})) \quad (1)$$

where $W_{syn} \in \mathbb{R}^{h \times 768}$ projects embeddings into a space optimized for similarity detection.

4.3.2 Antonym Space Projection

The antonym projection branch learns to highlight oppositional relationships:

$$\mathbf{x}_{ant} = \text{Dropout}(\text{ReLU}(W_{ant}\mathbf{x} + \mathbf{b}_{ant})) \quad (2)$$

This separation allows the model to distinguish between words that might appear in similar contexts but express contrasting meanings. The key insight is that antonyms should be similar in antonym space but dissimilar in synonym space.

4.3.3 Feature Fusion

The outputs from both branches are combined to create rich representations:

$$\mathbf{x}_{fused} = W_f[\mathbf{x}_{syn}; \mathbf{x}_{ant}] + \mathbf{b}_f \quad (3)$$

This fusion enables the model to consider both semantic perspectives simultaneously when making relationship predictions.

4.4 Graph-Based Relational Processing

For each word pair (w_1, w_2) , we construct a simple graph $G = (V, E)$ where $V = \{v_1, v_2\}$ and $E = \{(v_1, v_2), (v_2, v_1)\}$. The node features are the fused embeddings from the dual projections.

This graph representation might seem overly simple, but it becomes powerful when processed through transformer convolutions. The bidirectional edges ensure symmetric relationship detection, which is crucial since both synonymy and antonymy are symmetric relations.

4.5 Graph Transformer Convolutions

We apply transformer convolutions to capture higher-order semantic dependencies:

$$\mathbf{X}^{(l)} = \text{Dropout}(\text{ReLU}(\text{TransformerConv}(\mathbf{X}^{(l-1)}, E))) \quad (4)$$

The transformer convolution allows nodes to exchange information through attention mechanisms, enabling the model to focus on the most relevant semantic features for relationship determination.

4.6 Classification and Semantic Space Organization

After graph processing, we apply global mean pooling and classification:

$$\mathbf{x}_{pool} = \frac{1}{|V|} \sum_{i \in V} \mathbf{x}_i^{(L)} \quad (5)$$

$$\hat{y} = \sigma(\mathbf{W}_2 \text{Dropout}(\text{ReLU}(\mathbf{W}_1 \mathbf{x}_{pool} + \mathbf{b}_1)) + \mathbf{b}_2) \quad (6)$$

4.6.1 Margin-Based Loss for Semantic Organization

The margin loss is crucial for organizing the learned semantic spaces properly:

$$\mathcal{L} = \mathcal{L}_{BCE}(\hat{y}, y) + \lambda \mathcal{L}_{margin} \quad (7)$$

where:

$$\mathcal{L}_{margin} = \begin{cases} \max(0, m_{syn} - \tanh(\langle \mathbf{x}_{syn}^1, \mathbf{x}_{syn}^2 \rangle)) & \text{if } y = 0 \\ \max(0, \tanh(\langle \mathbf{x}_{ant}^1, \mathbf{x}_{ant}^2 \rangle) - m_{ant}) & \text{if } y = 1 \end{cases} \quad (8)$$

This loss guides the model to position synonym pairs with high similarity in synonym space ($> m_{syn} = 0.8$) and antonym pairs with high similarity in antonym space but low similarity in synonym space ($< m_{ant} = 0.2$).

4.7 Two-Stage Training Strategy

Our training approach follows a carefully designed two-stage process that leverages the strengths of both language-specific understanding and cross-lingual relationship modeling.

4.7.1 Stage 1: BERT Fine-tuning for Accelerated Learning

In the first stage, we fine-tune each language-specific BERT model for binary antonym classification. This serves as a warm-up phase that creates high-quality initial representations. The intuition is simple: before learning complex dual-space projections, the model should first understand what makes words semantically related in each language.

For each language l , we train the BERT model using standard classification loss:

$$\mathcal{L}_{BERT} = -\frac{1}{N_l} \sum_{i=1}^{N_l} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (9)$$

This fine-tuning typically converges within 3-5 epochs. The resulting models capture language-specific patterns for semantic relationship detection, providing strong initial embeddings for the subsequent dual encoder training.

4.7.2 Stage 2: Dual Encoder Training

In the second stage, we freeze the fine-tuned BERT encoders and train the dual encoder architecture. This separation is crucial: it allows the model to focus on learning optimal projection mappings without disrupting the already-optimized language representations.

The dual encoder trains for 10-20 epochs using the combined loss function, with careful learning rate scheduling (1×10^{-4}) to ensure stable convergence. The frozen BERT embeddings provide consistent, high-quality inputs while the dual projections and graph transformers learn to organize semantic relationships effectively.

4.7.3 Cross-Lingual Training Philosophy

Our approach balances universal and language-specific characteristics. While we use different BERT models for each language, the dual encoder architecture remains consistent across languages. This design recognizes that semantic relationship patterns may be universal (opposition vs. similarity), but their surface manifestations vary across languages.

Each language uses:

- Language-specific datasets from WordNet and ConceptNet
- Native BERT models that understand language-specific patterns
- Consistent architectural parameters that enable knowledge transfer
- Unified hyperparameters for fair cross-lingual comparison

This strategy ensures that the model benefits from both language-specific optimization and universal relationship modeling principles.

5 Experimental Setup

5.1 Datasets and Data Quality

Our evaluation uses real-world antonym datasets that represent a significant improvement over previous synthetic approaches. We source data from two authoritative repositories:

- **Open Multilingual WordNet (OMW):** Professional linguistic data providing high-quality antonym pairs across multiple languages
- **ConceptNet 5.5:** Large-scale semantic knowledge graph containing extensive multilingual relationship data [7]

The data collection process yields substantial datasets across our target languages:

Table 1: Dataset Statistics Across Languages

Language	WordNet Pairs	ConceptNet Pairs	Total Pairs
German	0*	2,678	2,678
French	571	5,703	6,095
Spanish	1,634	878	2,263
Italian	917	1,889	2,495
Portuguese	412	1,845	2,257
Dutch	298	1,567	1,865
Russian	156	1,123	1,279

*German WordNet data not available in OMW format.

This represents a 100-300x increase in dataset size and quality compared to previous synthetic approaches. The data undergoes careful preprocessing including language-specific character normalization and automatic train/validation/test splits (70

5.2 Implementation Details

Bhav-Net is implemented using PyTorch and PyTorch Geometric with the following configuration:

- **Architecture:** Hidden dimension 256, dropout rate 0.2, 2 graph transformer heads
- **Training:** Batch size 32, BERT learning rate 2×10^{-5} , dual encoder learning rate 1×10^{-4}
- **Loss:** Margin thresholds $m_{syn} = 0.8$, $m_{ant} = 0.2$, margin weight $\lambda = 0.5$
- **Optimization:** Adam optimizer with automatic mixed precision for efficient GPU training

All experiments use early stopping based on validation performance to prevent overfitting.

5.3 Evaluation Protocol

We evaluate using standard classification metrics: accuracy, precision, recall, and F1-score. Additionally, we analyze the learned semantic spaces through visualization of projection similarities and margin loss convergence patterns to ensure the model learns interpretable representations.

6 Results

Table 2 presents the performance of Bhav-Net across all seven target languages. The results demonstrate consistent effectiveness across diverse linguistic contexts, validating our architectural design choices.

Table 2: Bhav-Net Performance Across Languages

Language	Accuracy	Precision	Recall	F1-Score
German	—	—	—	—
French	—	—	—	—
Spanish	—	—	—	—
Italian	—	—	—	—
Portuguese	—	—	—	—
Dutch	—	—	—	—
Russian	—	—	—	—
Average	—	—	—	—

7 Analysis

7.1 Why Bhav-Net Works Across Languages

The cross-lingual effectiveness of Bhav-Net stems from its ability to capture universal semantic principles while respecting language-specific characteristics. Several factors contribute to this success:

7.1.1 Universal Semantic Patterns

The dual-space projection mechanism models fundamental patterns of opposition and similarity that transcend linguistic boundaries. Humans across cultures understand concepts like "hot" vs "cold" or "big" vs "small" regardless of their native language. Bhav-Net captures these universal patterns in its architectural design.

7.1.2 Language-Specific Optimization

Using dedicated BERT models for each language ensures that the model understands language-specific nuances. German compound words, French gendered nouns, Spanish verb conjugations, and Russian case systems all carry semantic information that might be lost in a generic multilingual approach.

7.1.3 Relational Reasoning Through Graphs

The graph transformer architecture enables reasoning about relationships rather than just individual words. This mirrors how humans understand semantic relationships - we don't just compare words in isolation but consider them within their broader semantic context and connections.

7.2 Architectural Analysis: Modeling Human Semantic Understanding

Bhav-Net's architecture closely parallels how humans process semantic relationships:

7.2.1 Dual Cognitive Frameworks

The dual projection mechanism reflects how humans maintain separate cognitive frameworks for understanding similarity versus opposition. When we encounter the words "warm" and "hot," we simultaneously recognize their similarity (both relate to high temperature) and their degree of difference (hot is more intense than warm). Similarly, with "hot" and "cold," we recognize both the shared domain (temperature) and the oppositional relationship.

7.2.2 Contextual Integration

The graph-based processing mirrors how humans integrate contextual information when determining relationships. We don't make relationship judgments based on isolated word comparisons but consider the broader semantic context and connections between concepts.

7.2.3 Attention and Focus

The transformer convolutions enable selective attention to relevant semantic features, similar to how humans focus on specific aspects of word meaning when determining relationships. The attention mechanism allows the model to weight the most informative features for each language and relationship type.

7.3 Semantic Space Organization

Analysis of the learned semantic spaces reveals meaningful organization that aligns with linguistic intuition:

7.3.1 Synonym Space Characteristics

In the synonym space, semantically related words cluster together with clear boundaries corresponding to semantic fields. Words like "happy," "joyful," and "cheerful" form tight clusters, while maintaining appropriate distances from unrelated concepts.

7.3.2 Antonym Space Characteristics

The antonym space develops a different organizational principle where oppositional pairs exhibit high similarity. Pairs like "hot-cold," "big-small," and "happy-sad" cluster together in this space, indicating that the model has learned to recognize oppositional patterns across different semantic domains.

7.3.3 Cross-Lingual Consistency

Remarkably, these organizational principles remain consistent across languages. French "chaud-froid" (hot-cold), German "groß-klein" (big-small), and Spanish "feliz-triste" (happy-sad) all exhibit similar patterns in their respective spaces, suggesting that Bhav-Net captures universal aspects of semantic opposition.

7.4 Training Dynamics and Convergence

The two-stage training approach proves crucial for stable learning. The BERT fine-tuning stage provides strong initial representations that accelerate dual encoder convergence. Without this warm-up phase, the dual projections struggle to organize effectively, leading to suboptimal performance.

The margin loss successfully enforces the desired semantic organization, with convergence patterns showing clear separation between relationship types across all languages. The margin

thresholds ($m_{syn} = 0.8$, $m_{ant} = 0.2$) create decision boundaries that align with human intuitions about semantic relationships.

7.5 Generalization and Transfer

The consistency of performance across languages with varying dataset sizes (from 1,279 pairs in Russian to 6,095 in French) demonstrates the robustness of our approach. The architectural principles generalize well even to languages with limited training data, suggesting that the fundamental patterns of semantic opposition are indeed universal.

8 Conclusion

This paper introduced Bhav-Net, a multilingual architecture for semantic relationship detection that successfully bridges the gap between universal semantic principles and language-specific characteristics. Our approach demonstrates that with thoughtful architectural design, it's possible to create models that work effectively across diverse linguistic contexts.

8.1 Key Contributions

We make three primary contributions to the field of computational semantics:

Dual-Space Architecture: The dual projection mechanism provides a principled approach to modeling different types of semantic relationships. By maintaining separate spaces for synonyms and antonyms, Bhav-Net captures the distinct distributional patterns exhibited by these relationship types.

Multilingual Effectiveness: Our comprehensive evaluation across seven languages demonstrates that semantic relationship detection can be effectively generalized across linguistic boundaries while respecting language-specific nuances.

Intuitive Design Philosophy: Bhav-Net's architecture reflects how humans process semantic relationships - maintaining separate cognitive frameworks for similarity and opposition while integrating contextual information through relational reasoning.

8.2 Practical Impact

The practical implications of this work extend beyond academic interest. Multilingual semantic understanding is crucial for global applications including machine translation, cross-lingual information retrieval, and multilingual content generation. By providing robust antonym detection across languages, Bhav-Net enables more nuanced semantic processing in multilingual systems.

The model's ability to work with limited training data (as demonstrated by the Russian experiments with only 1,279 pairs) makes it particularly valuable for low-resource languages where comprehensive semantic resources are scarce.

8.3 Limitations and Future Directions

While Bhav-Net demonstrates consistent performance across the tested languages, several limitations suggest directions for future work:

Coverage of Relationship Types: Current work focuses on antonymy and synonymy, but human semantic understanding encompasses a broader range of relationships including hyponymy, meronymy, and causal relationships. Extending the dual-space approach to handle multiple relationship types represents a natural progression.

Contextual Sensitivity: The current model treats word pairs in isolation, but semantic relationships can be context-dependent. Future work could integrate contextual information to handle cases where words exhibit different relationships in different contexts.

Computational Efficiency: The dual encoder architecture, while effective, requires training and maintaining separate models for each language. Exploring approaches that balance cross-lingual sharing with language-specific adaptation could improve computational efficiency.

8.4 Broader Implications

Bhav-Net’s success suggests that cognitive-inspired architectures can provide both performance benefits and interpretability advantages. The dual-space design reflects human cognitive patterns while enabling computational implementation - a promising direction for future semantic modeling research.

The consistent cross-lingual performance also supports the hypothesis that fundamental semantic patterns are universal, even though their surface manifestations vary across languages. This finding has implications for both computational linguistics and cognitive science.

8.5 Final Thoughts

Semantic understanding remains one of the most challenging aspects of natural language processing. By combining insights from cognitive science, linguistic theory, and modern deep learning, Bhav-Net demonstrates that thoughtful architectural design can lead to both effective and interpretable solutions.

As multilingual AI systems become increasingly important in our connected world, approaches like Bhav-Net that respect both universal principles and linguistic diversity will become essential. The path forward lies not just in scaling existing approaches, but in developing architectures that reflect the complexity and beauty of human semantic understanding.

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